The Lookout and the Captain: Mode Conditioning and Event-Triggered Decoding in Frozen Vision-Language Models

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Abstract

A small vision model can select a behavioral mode for a larger model and decide when that model should run. We study these two functions separately in frozen vision-language models with nominal sizes of 450M, 3B, and 27B parameters. A learned 256-dimensional packet encodes an operator's mode instruction. State-conditioned receivers inject the packet into three language-model layers; the larger model also receives the image. The task has three possible actions: greet, stop, and proceed. On held-out photographs, the packet changes action selection, with substantial variation across modes and adapter seeds. At 27B, its increase over a constant adapter is 0.644, 0.867, and 0.467 for the three modes. Detector-controlled packet selection incurs errors that can be traced to the selected packet on each image. In a replay of short image streams, waking the model on changes in the detector's verdict avoids 83.5% of model calls, with 98.1% social-mode and 99.0% caution-mode agreement with decoding every frame at 27B. This is an invocation result, not a measured energy saving or a delay guarantee. Rendered-scene tests expose failures that depend on the adapter seed, instruction phrase, training domain, and constant-adapter baseline. The results support mode conditioning and selective invocation in this restricted task. They do not establish general instruction robustness, robot-control safety, or transfer across arbitrary visual domains.

1 Introduction

An image can determine both what a model should do and whether it needs to run again. These are different decisions. A person detector can report that someone is present without determining whether a robot should greet that person or stop. An operator supplies that policy. Similarly, a change in pixels need not require a new action, while a small change that alters the task state may require one.

We examine a two-model arrangement. The *lookout* is a small person classifier or detector. The *captain* is a frozen vision-language model that receives the camera image and produces an action. An external policy maps the detector's binary verdict to a mode packet: use the operator's social or caution mode when a person is detected, and explore mode otherwise. A second rule decides whether to invoke the captain or retain its previous action. The packet represents a mode instruction. In the main experiments it does not encode the detector's visual features or describe an object.

This construction lets us ask four empirical questions. Does the packet change action selection beyond a separately trained constant adapter? How do detector errors change that effect? How does conditioning behave under contrary text, missing images, and a new rendering domain? Finally, how often can the model reuse an action while matching a model invoked on every frame?

The evidence is a controlled case study, not a general robot benchmark. We use three frozen hosts, three adapter seeds for the main comparisons, fixed photographic evaluation sets, and matched interventions on each image. The larger host demonstrates that the mechanism operates beyond a small language model. The two model families differ in architecture and pretraining, so their comparison is not a controlled scaling law. Negative results are part of the answer: some modes barely improve over the constant adapter, some held-out instruction phrases fail, and mixed visual-domain training does not give a uniformly better adapter.

2 Relation to existing methods

Prefix tuning learns continuous activations that guide a frozen language model [5]. Activation engineering constructs steering directions from activation differences between prompts [11], while representation engineering studies control and monitoring through model representations [13]. Our packet and receivers are trained components, and each receiver reads the target model's current residual state. This differs from adding a single fixed steering direction, but does not make the broader idea of activation-based conditioning new.

Hypernetworks generate the weights of another network [2]. The implementation here instead generates a latent packet and residual updates; it does not generate LoRA matrices or alter the host's stored weights. We therefore describe the actual emitter and receivers rather than treating all conditional modules as the same architecture.

Selective invocation also differs from speculative decoding [4]. Speculative decoding uses a smaller model to propose tokens that a larger model verifies. Here the detector decides whether to skip a whole model invocation and retain the last action. There is no token-level verification or exact output-distribution guarantee. Our contribution is the combined experimental analysis of these mode-selection and invocation decisions, including controls that locate their failures.

3 Method

3.1 Task, hosts, and photographic data

The model answers a fixed request for a JSON object whose action is `greet`, `stop`, or `proceed`. The reference policy is: social mode greets a present person; caution mode stops for a present person; explore mode always proceeds; every mode proceeds on an absent image. These are task labels, not executable robot actions.

The hosts are LiquidAI/LFM2.5-VL-450M, LiquidAI/LFM2.5-VL-3B, and Qwen/Qwen3.8-27B [7, 8, 10]. Their measured language towers have 16, 30, and 64 layers, with hidden widths 1024, 2048, and 5120. All host parameters are frozen. CUDA runs use bfloat16 host computation and float32 adapter computation. Qwen's thinking mode is disabled. Model identifiers, available revision pins, and checkpoint hashes accompany the code.

We draw photographs from COCO val2017 [6]. A present image has at least one non-crowd person annotation whose `area`/image-area is at least 0.05. This uses the annotation's area, not its bounding-box area; the historical source labels that field imprecisely. An absent image has no person annotation. Other images are excluded. The eligible pools contain 1505 present and 2307 absent images. Sampling uses sorted image IDs and a fixed Python random seed of 42.

The adapter training set contains 40 present and 40 absent photographs. The initial evaluation set contains a disjoint 30 present and 30 absent photographs. Detector-switch and box experiments use a further disjoint 100 present and 100 absent photographs. The remaining 3472 eligible photographs train the small classifier and provide detector-size calibration. Each evaluation set is shared across host and adapter seeds. Reusing it for subsequent design choices makes the follow-ups exploratory evaluations on a reused set, rather than new independent confirmation. COCO val2017 is also a standard detector validation set; this study cannot establish that it is unseen by the pretrained hosts.

3.2 A packet and three state-conditioned receivers

There are ten instruction phrasings per mode. Positions 0–7 train the adapter; positions 8 and 9 are held out. For each phrase, we run a text-only exchange containing the mode assignment and an assistant acknowledgment, and extract the last-position hidden state at layer round$(0.79L)$. The packet source therefore contains the mode text, not image features.

Let s be that source state. A mean and scalar scale fitted on the 24 training-phrase states center and normalize s, followed by RMS normalization. The emitter is a linear map and GELU:

$$z = \mathrm{GELU}(W_E\ \mathrm{RMSNorm}((s-\mu)/\sigma) + b_E), \qquad z \in \mathbb{R}^{256}. \tag{1}$$

At each injection layer ℓ, the receiver combines projections of z, the RMS-normalized current residual $h_{\ell,t}$, and a learned position embedding e_t. Packet and residual projections each have width 128, and the position embedding

has width 64. A 320-to-512-to-d MLP with GELU produces $u_{\ell,t}$, where d is the host width. The receiver reads a detached residual state; training still propagates through the frozen host operations to the adapter.

$$h'_{\ell,t} = h_{\ell,t} + \alpha \|h_{\ell,t}\|_2 \frac{u_{\ell,t}}{\max(\|u_{\ell,t}\|_2, 10^{-6})}. \tag{2}$$

The three injection layers are the rounded depth fractions $(0.29, 0.46, 0.63)$, clamped to valid layers and deduplicated. The resulting indices are $(5, 7, 10)$, $(9, 14, 19)$, and $(19, 29, 40)$; source indices are 13, 24, and 51. These indices were not selected by a layer sweep. Injection begins at the last prompt position and continues during decoding. There are 16 learned position embeddings, and later positions use the last embedding. The operating dose is $\alpha = 0.15$.

The receivers have separate parameters at each layer and share the packet. Because they read image-conditioned host states, a packet that is independent of the image can still produce an image-dependent intervention. Its text-only source does not imply domain-independent behavior.

The emitter and three receivers contain approximately 2.83M, 5.06M, and 11.75M trainable parameters for the respective hosts, excluding the 771-parameter auxiliary training head. The packet width remains 256 while the receiver output width follows the host. This parameter count describes the complete emitter and receiver bank.

3.3 Training and comparison arms

We train each adapter for 192 optimizer steps with eight sequentially accumulated examples per step, AdamW at learning rate 0.003, no weight decay, and adapter gradient-norm clipping at 1.0. Each example draws a mode, a training image, and, for the mode adapter, a training phrase. The loss is teacher-forced cross entropy on the reference JSON action and EOS token. A three-class auxiliary head on the mode packet adds a cross-entropy term of weight one. The head is trained but not used at evaluation. Its parameters are optimized alongside the adapter; the historical clipping call covers the adapter parameters only.

A separate *constant adapter* uses the mean training source state for every mode and has no auxiliary mode head. Its image-to-action training targets still span the three modes. It is therefore a learned baseline with its own action preferences, not the unmodified frozen model. The mode and constant adapters also consume different random draws in training. The comparison measures the effect of the complete trained conditioning arrangement; it does not isolate the auxiliary loss or equalize every training choice.

The four main arms are:

- **Packet:** the mode adapter receives the requested mode packet; the prompt contains no mode instruction.
- **Swapped:** the same adapter receives the next mode in a fixed social–caution–explore cycle. This is a deterministic wrong-mode control, called `shuffled` in the saved records.
- **Constant:** the constant adapter receives its fixed source; the prompt contains no mode instruction.
- **Text:** the constant adapter remains active and the prompt contains the requested mode instruction.

Thus Packet versus Text compares two trained adapter arrangements. It does not compare packet conditioning with an untouched model's best possible prompt. There is no broad prompt search.

The main seeds are 42, 1042, and 2042. Before evaluation, recorded checks require unchanged sampled host parameters, identical next-token logits at zero injection dose on a test image, and at least 0.90 teacher-forced token accuracy on the held-out phrases with eight training images. All main adapter pairs pass. The parameter check samples parameters, and the phrase check includes predictable JSON syntax; neither is a complete weight-integrity or held-out action-accuracy test.

3.4 Evaluation and uncertainty

Decoding is greedy with a maximum of 24 new tokens. The parser selects the first valid JSON object containing an allowed action, otherwise the first allowed action word, and otherwise `other`. We replay this parser on every saved individual-image response in the audit. Stream records retain the action and parse path, rather than the full generated response.

For mode m, let $p_A(m)$ be the fraction of present images on which arm A returns the mode's target action. We report the raw shares and paired differences, especially $p_{\text{Packet}} - p_{\text{Constant}}$. On absent images, greet or stop is a task error; proceed is correct. A high agreement with another model is reported separately from reference-policy correctness.

Means give each tested adapter seed equal weight. Intervals are post-hoc, pointwise 95% percentile bootstrap intervals with 10,000 replicates. We resample image IDs within a label, retaining all paired arms and the tested seeds for each image. For rendered data, a within-label observation is one scene draw. For streams, we resample complete 16-step episodes. We do not treat repeated seeds, arms, or frames in an episode as independent observations.

These intervals describe sampling variation conditional on the tested seeds, fixed phrase assignment, and source population. They do not cover an unseen seed distribution, new instruction phrasings, or adaptive experiment selection. We also give per-seed results. At a boundary, percentile intervals can collapse to a point; an empirical zero error with such an interval does not prove zero population error. The historical plan contains dated questions, thresholds, amendments, and readings. Those internal records document the exploratory sequence; they are not an external preregistration, and their reading bands are not statistical significance tests.

4 Conditioning on photographs

Table 1 and Figure 1 show clear mode-dependent effects, but no uniform benefit in every host and mode. Caution improves substantially on all three hosts. Social mode on 3B has a mean improvement of only 0.022, and explore mode on 450M improves by 0.044. Both intervals include zero. At 27B, Packet exceeds Text by 0.111, 0.389, and 0.333 for social, caution, and explore respectively, under the particular Text control defined above.

Table 1: Target-action share on 30 present photographs, averaged over three adapter seeds. Brackets give paired image-bootstrap intervals. Constant is the separately trained constant adapter.

Host	Mode	Constant	Packet	Text	Packet–Constant	Packet–Text
450M	Social	0.300	0.633	0.967	0.333 [0.144, 0.511]	-0.333 [-0.489, -0.178]
450M	Caution	0.222	1.000	0.922	0.778 [0.700, 0.856]	0.078 [0.033, 0.133]
450M	Explore	0.478	0.522	0.533	0.044 [-0.089, 0.178]	-0.011 [-0.144, 0.122]
3B	Social	0.467	0.489	0.922	0.022 [-0.211, 0.256]	-0.433 [-0.600, -0.256]
3B	Caution	0.089	0.756	0.911	0.667 [0.567, 0.756]	-0.156 [-0.256, -0.033]
3B	Explore	0.444	1.000	0.967	0.556 [0.433, 0.678]	0.033 [0.000, 0.078]
27B	Social	0.333	0.978	0.867	0.644 [0.567, 0.722]	0.111 [0.011, 0.222]
27B	Caution	0.133	1.000	0.611	0.867 [0.789, 0.933]	0.389 [0.289, 0.489]
27B	Explore	0.533	1.000	0.667	0.467 [0.389, 0.533]	0.333 [0.256, 0.411]

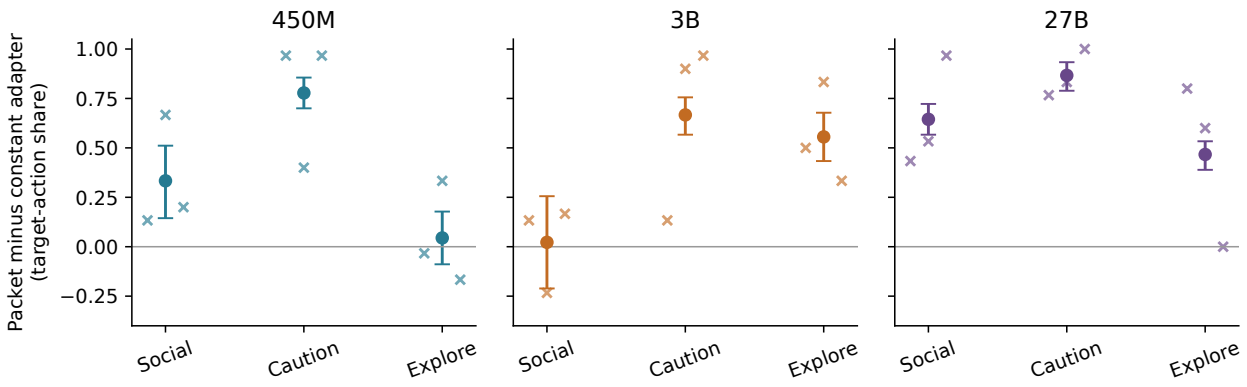


Figure 1: Conditioning effects on present photographs. Circles and error bars show means and image-bootstrap intervals conditional on the three seeds; crosses show the three separate seed means.

Wrong-mode packets change the actions, including when Packet and Constant have similar means. This is evidence that packet content matters, although its effect can be poorly aligned with the requested task. Absent-image behavior must be examined separately: 27B Packet never greets in social mode on the 30 absent photographs, but its mean caution stop share is 0.044. Table 6 reports both the wrong-mode contrast and absent-image shares. The findings establish restricted action conditioning rather than general multimodal generation or reliable obstacle avoidance.

5 Replacing the annotation with a detector

5.1 Detector selection and measured error

The external switch chooses the requested social or caution packet when its input verdict is present, and the explore packet otherwise. An oracle switch uses the COCO label. A learned classifier uses ImageNet-pretrained MobileNetV3-small features [3] with a two-class linear head trained for three epochs, seed 42, on the 3472-image remainder. An off-the-shelf YOLO26 detector [12] uses a person box with confidence at least 0.25 and bounding-box fraction at least 0.05 as its present rule. The box rule and COCO annotation-area rule are not identical.

Table 2 gives the detector results. Detector size was chosen by the fewest misses on the separate calibration remainder, breaking ties toward the smaller detector. The medium detector has the fewest calibration misses. Both small and medium detectors have zero misses on the 100 present evaluation photographs, but still have false alarms. This is a finite-set observation, not proof of a perfect detector.

Table 2: YOLO26 detection results. Evaluation has 100 present and 100 absent images; calibration has 1335 present and 2137 absent images. Timing is the recorded batch-inference wall time per calibration image, including the prototype's processing overhead.

Detector	Params (M)	Eval misses	Eval false alarms	Calib. misses	Calib. false alarms	ms/image
YOLO26N	2.41	5	2	25	25	3.49
YOLO26S	9.50	0	1	12	22	3.58
YOLO26M	20.41	0	2	10	22	4.43

Table 3: Detector minus oracle target-action share on the 100 present evaluation images, with paired intervals across images and means over three seeds. Small and medium YOLO host runs were completed on the two LFM hosts; 27B uses the nano detector.

Host	Detector	Social gap	Caution gap
450M	MobileNet	-0.053 [-0.093, -0.020]	-0.077 [-0.127, -0.033]
3B	MobileNet	-0.037 [-0.077, -0.007]	-0.073 [-0.117, -0.037]
450M	YOLO26n	-0.027 [-0.060, 0.000]	-0.037 [-0.073, -0.007]
3B	YOLO26n	-0.013 [-0.033, 0.000]	-0.033 [-0.067, -0.007]
27B	YOLO26n	-0.033 [-0.070, -0.007]	-0.047 [-0.090, -0.010]
450M	YOLO26s	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
3B	YOLO26s	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
450M	YOLO26m	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
3B	YOLO26m	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

The nano detector produces measurable gaps, including at 27B (Table 3). The zero present-image gaps for small and medium YOLO follow from selecting the same packet as the oracle on every present evaluation image. They do not establish behavioral equivalence on arbitrary images or on the absent class.

5.2 An error-conditioned identity

The switch uses the same frozen host and adapters as the oracle arm. Let E_i indicate a detector error on image i, and let $C_i = I(a_i^O = a_m) - I(a_i^D = a_m)$ be the difference in target-action indicators between the oracle and

detector-selected arms. When the detector is correct, the selected packet is identical. The saved greedy actions agree in these cells. It follows that

$$p_O - p_D = \frac{1}{n}\sum_i E_i C_i = \widehat{\Pr}(E)\,\overline{C}_{\{E=1\}}, \qquad |p_O - p_D| \le \widehat{\Pr}(E). \tag{3}$$

The error-conditional cost is undefined when there are no errors, in which case the gap is zero. We verify the identity per seed and label. The flipped-verdict arm inverts the detector's verdict, not the annotation. On an actual detector error, it therefore selects the oracle packet and allows the corresponding action to be checked.

The identity is a bookkeeping consequence of a deterministic switch. It does not justify multiplying the error rate by a forced-flip cost averaged over all images. For example, the 27B seed-42 caution run has a 0.05 miss rate, a 0.94 population-wide oracle-minus-flipped share, and a 0.05 oracle-minus-detector gap. The product $0.05 \times 0.94 = 0.047$ is not an upper bound on that gap. Errors can occur on precisely the images where the action cost is greatest.

6 Contrary text and additional visual input

6.1 Contrary instructions

The contrary-text experiments add a direct request for the next mode in the same fixed cycle. They include Packet alone, contrary text with the constant adapter, Packet plus contrary text, Packet plus congruent text, and swapped Packet plus contrary text. We evaluate the original dose and 1.5 times that dose. Table 7 reports the direct difference between Packet plus contrary text and contrary text with the constant adapter.

This effect is substantial for all three 27B modes at the original dose, but results on the LFM hosts vary by seed and mode. The historical *ownership* ratio divides this effect by Packet–Constant, suppressing denominators below 0.05 in magnitude. That ratio can exceed one, reverse sign, or become unstable when the baseline contrast is small. We retain it in the audited records but use raw shares and paired differences as the primary presentation. The test uses two fixed held-out phrasings, not an adversarial search, and provides no prompt-injection security guarantee.

Increasing dose changes absent-image behavior as well as contrary-text responses. A targeted 3B seed-42 follow-up instead doubles training to 384 steps at the original dose. Its caution Packet share rises to 1.000, and its Packet-plus-contrary minus contrary-text effect is 0.733. Its constant-adapter caution share also changes to 0.500. This is a useful successful intervention on one selected seed, but does not isolate training duration from the resulting baseline change or establish a universal dose-versus-training tradeoff.

6.2 Box features and a blank host image

A separate variant adds a learned projection of eight detector features to the normalized mode source: presence, clipped count, largest-box fraction, best qualifying confidence, and the largest box's center and dimensions. The mode adapter remains frozen. Only a zero-initialized linear projection trains, for 192 steps at learning rate 0.0003. This is the one experiment family in which continuous detector features enter the packet. The recorded plan documents an earlier unstable formulation and the amended normalization and learning rate.

The comparison includes correct boxes, a detector-controlled mode switch, deliberately false boxes, and the constant adapter. False boxes put a present image's boxes on absent images and zeros on present images. Box features improve 3B caution action share over the binary switch by 0.223 on present photographs, with smaller or negative effects in the other host-mode combinations (Table 8).

In a further test, the host receives a uniform black 640-by-480 image during projection training and evaluation, while the detector still sees the photograph. Most seeds then produce few person-directed actions. One 3B caution seed is different: the box arm stops on every present image and the deliberately false-box arm also stops on every absent image. The contrast shows that access to the original image can matter, and that a successful input channel can transmit incorrect information. It does not define a safe operating dose or prove that visual grounding always overrides a false packet. No 27B box or blank-image run is included.

7 When should the captain run?

7.1 Replay design

We construct 60 episodes of 16 steps from the 200-image switch evaluation pool, using a common seed of 2026. Each episode starts with a sampled photograph. Later steps change the base photograph with probability 0.2 and otherwise hold it. Every step receives a random 94–100% crop and brightness multiplier in $[0.96, 1.04]$. The realized stream uses 144 distinct base photographs and contains 227 cuts including 60 episode starts. Its pixel manifest is identical across hosts and seeds.

Every gate invokes the model at episode start. After that: *every frame* always invokes; *episode start* never invokes; *verdict* invokes when the person verdict differs from that at the last invocation; *boxes* also reacts to changes in count, area, or center; *pixels* uses a fixed 0.04 mean absolute pixel-difference threshold; and *scene cuts* invokes on the known base-image changes. Otherwise the last action is retained.

The replay caches the greedy model response for a frame and selected packet, then applies these gate rules. It measures the fraction of invocations the policy would avoid and the resulting action sequence. It is not a live robot trial. Scene cuts are a useful reference gate, not an agreement ceiling: crop and brightness changes can alter the every-frame model's response between cuts.

7.2 Invocation savings and task correctness

The verdict gate invokes on 158 of 960 steps and avoids 83.5% of calls. It gives high agreement with every-frame decoding at all three host sizes, but the reference-policy accuracy of the smaller hosts can remain substantially lower than their agreement (Table 4). Matching a fallible model does not make its actions correct.

Table 4: Verdict-gate replay. Means use three adapter seeds on the same 60 episodes; brackets resample complete episodes. Correctness change is verdict minus every-frame decoding against the reference policy.

Host	Mode	Agreement	Reference correctness	Correctness change
450M	Social	0.972 [0.956, 0.986]	0.786 [0.712, 0.857]	-0.007 [-0.022, 0.006]
450M	Caution	0.988 [0.974, 0.998]	0.970 [0.943, 0.992]	-0.008 [-0.022, 0.002]
3B	Social	0.964 [0.930, 0.986]	0.707 [0.611, 0.797]	-0.007 [-0.040, 0.015]
3B	Caution	0.986 [0.972, 0.997]	0.869 [0.836, 0.903]	-0.005 [-0.019, 0.006]
27B	Social	0.981 [0.964, 0.994]	0.957 [0.930, 0.980]	-0.002 [-0.018, 0.014]
27B	Caution	0.990 [0.978, 1.000]	0.976 [0.953, 0.994]	0.001 [-0.011, 0.011]

The 27B decode helper records a mean 0.453 seconds per actual decode. This timing is specific to the GPU, cached preprocessing, prompt, and short response used here. The experiment does not measure complete system energy, detector-plus-host latency, or production throughput. Detector work continues on skipped frames. Multiplying avoided calls by this decode time is an estimate of avoided model work, not an end-to-end saving measurement. Likewise, the poor savings of this one pixel threshold under deliberate pixel jitter do not establish that semantic gates are always necessary.

7.3 Delay includes unresolved cuts

Time from a scene cut to the next invocation is not the same as time to a necessary action correction: some scene changes require no different action. Even as a descriptive responsiveness measure, it must include cuts whose next invocation is not observed.

There are 167 cuts after episode start. Under the verdict gate, 130 have an observed later or same-step invocation, and 37 remain unresolved when the episode ends. The observed delays have mean 1.44 steps and maximum 12 steps; 33 exceed one step. The unresolved cases are right-censored, not zero-delay successes. The historical 0.984-step average included the 60 zero-delay episode starts and excluded unresolved cuts. It cannot support a one-frame response claim. Table 9 reports all gates using the correct post-start denominator; the analysis also supplies each cut's event or censoring record.

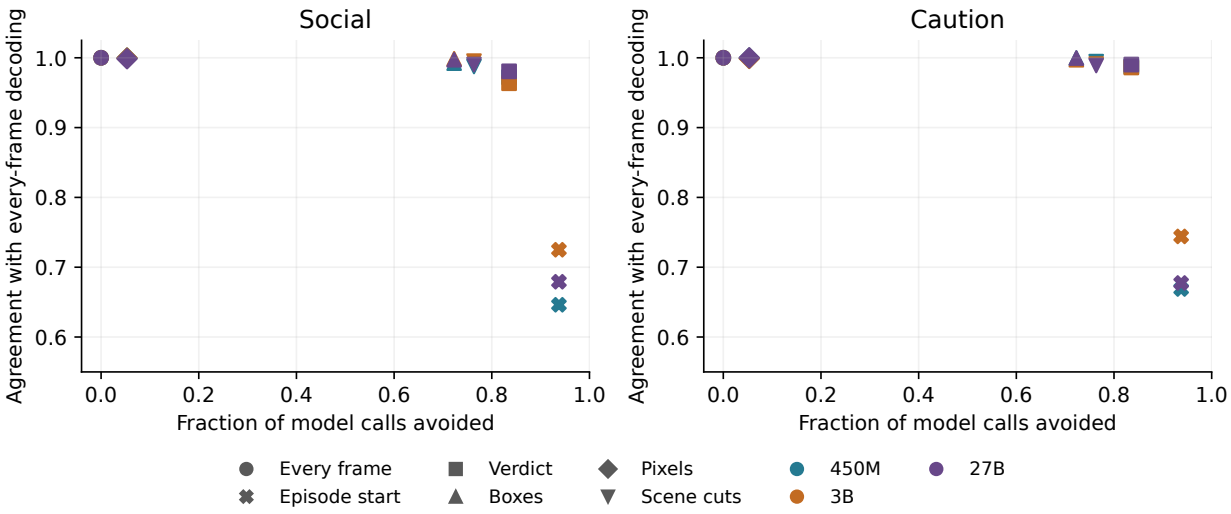


Figure 2: All six invocation policies, averaged over three seeds. The fixed pixel threshold invokes on most jittered frames; the verdict and box rules avoid more calls in this replay. The figure does not compare tuned pixel gates or measured energy.

8 Transfer to rendered scenes

8.1 Changing the visual domain

We render paired present and absent scenes in MuJoCo [9]. Each pair shares camera, room, and lighting draws. The study first uses capsule-based human geometry, then one textured human scan from Digital Reality Lab's HSRD-100 collection [1]. Evaluation has 100 scene draws, each rendered with and without the person. The human identity and pose are fixed; rooms, camera placement, and illumination vary.

Two pixel versions survive in the records. Their scene draws are identical, but their PNG hashes differ. The original LFM transfer runs and the 3B training follow-ups use the first version; the 27B runs and later 450M follow-ups use the second. We pair arms within each run. The mixed-versus-fixed-baseline comparisons below use identical pixels and mode checkpoints. Comparisons across pixel versions cannot attribute every change to adapter training alone. Host transfer tests use requested-mode packets as in the original conditioning experiment; detector performance is evaluated separately.

All three YOLO sizes miss all 100 present capsule scenes at each of the two tested capsule appearance settings. On the scan-based scenes, the nano detector finds 99 of 100 present scenes with no false alarms; the medium detector has no errors on this set. The change establishes that this particular representation of the person matters for these detectors. It does not separate all effects of mesh geometry, texture, and rendering, or establish a general simulator-to-real transfer result. Indeed, the tested direction is photographs to rendered scenes.

A 3B diagnostic with direct caution text has a 0.267 difference between present and absent stop shares on a 60-image capsule subset. The source code uses the frozen host without an adapter for that diagnostic Text arm, unlike the main Text control. The result is partial discrimination, not evidence that every host perceives the capsule reliably.

8.2 Training domain and baseline choice

We compare the original photograph-trained adapters with mixed training on the 80 photographs plus 40 present and 40 absent rendered training scenes. Rendered training uses a separate draw seed and scene-index range. The doubled-budget mixed recipe uses 384 steps. Most follow-ups use all three seeds at 450M and 27B; the 3B mixed variants have only seeds 42 and 1042. These are targeted follow-ups, not a complete factorial design.

Table 5 separates raw Packet behavior from the constant baseline. In the mixed recipe, retraining the constant adapter can make it strongly favor a person-directed action even without mode input. A small Packet–Constant

contrast can therefore reflect a changed baseline, a weak mode adapter, or both.

The August 27 follow-up pairs the mixed mode adapter with the original photograph-trained constant adapter. The Packet action sequence stays the same; only the comparison baseline changes. At 27B, the mean social and caution contrasts become 0.950 and 0.820 on rendered present scenes. Explore has a small contrast because the original constant adapter already proceeds on most rendered scenes. At 450M, the same analysis exposes a real explore failure: Packet proceeds on only 0.167 of present scenes while the original constant adapter proceeds on all of them. The original constant weights are reused, but their Text control is evaluated again. Across the two pixel versions, eight Text actions differ in the 450M comparisons; the constant, Packet, and swapped-Packet action sequences used for the fixed-baseline checks agree.

Table 5: Rendered present scenes (100 draws). Photo: original training; Mixed: mixed-domain mode and constant adapters, 384 steps; Mixed/fixed: the same mixed mode adapter with the original constant baseline. The Packet column is unchanged between the two mixed rows. Brackets give within-label scene-bootstrap intervals.

Host	Recipe	Mode	Seeds	Constant	Packet	Packet–Constant
450M	Photo	Social	3	0.000	0.467	0.467 [0.373, 0.563]
450M	Photo	Caution	3	0.000	0.690	0.690 [0.673, 0.707]
450M	Photo	Explore	3	1.000	0.833	-0.167 [-0.200, -0.133]
450M	Mixed	Social	3	0.263	0.500	0.237 [0.137, 0.337]
450M	Mixed	Caution	3	0.253	0.833	0.580 [0.517, 0.643]
450M	Mixed	Explore	3	0.483	0.167	-0.317 [-0.387, -0.250]
450M	Mixed/fixed	Social	3	0.000	0.500	0.500 [0.400, 0.600]
450M	Mixed/fixed	Caution	3	0.000	0.833	0.833 [0.800, 0.867]
450M	Mixed/fixed	Explore	3	1.000	0.167	-0.833 [-0.867, -0.800]
3B	Photo	Social	3	0.000	0.060	0.060 [0.023, 0.103]
3B	Photo	Caution	3	0.000	0.410	0.410 [0.383, 0.440]
3B	Photo	Explore	3	1.000	1.000	0.000 [0.000, 0.000]
3B	Mixed	Social	2	0.500	0.500	0.000 [-0.100, 0.100]
3B	Mixed	Caution	2	0.070	1.000	0.930 [0.895, 0.960]
3B	Mixed	Explore	2	0.430	1.000	0.570 [0.540, 0.605]
27B	Photo	Social	3	0.030	0.623	0.593 [0.560, 0.623]
27B	Photo	Caution	3	0.013	0.623	0.610 [0.547, 0.673]
27B	Photo	Explore	3	0.957	1.000	0.043 [0.020, 0.067]
27B	Mixed	Social	3	0.357	0.980	0.623 [0.583, 0.663]
27B	Mixed	Caution	3	0.347	0.833	0.487 [0.440, 0.533]
27B	Mixed	Explore	3	0.297	1.000	0.703 [0.660, 0.743]
27B	Mixed/fixed	Social	3	0.030	0.980	0.950 [0.927, 0.973]
27B	Mixed/fixed	Caution	3	0.013	0.833	0.820 [0.790, 0.853]
27B	Mixed/fixed	Explore	3	0.957	1.000	0.043 [0.023, 0.067]

Some failures coincide exactly with held-out phrase assignment. For the mixed 27B seed-42 caution adapter, the target-action shares on rendered present scenes are 0.000 for phrase 8 and 1.000 for phrase 9. At 450M, the mixed social adapter shows that same 0.000/1.000 split at all three seeds. The evaluation alternates the phrases across scene IDs rather than applying both phrases to every image, so this is a descriptive association. It is strong evidence against assuming broad paraphrase robustness; it does not by itself identify the causal contribution of phrase and scene subset.

Crossing initialization and batch-order seeds in selected mixed training runs changes both Packet and Constant behavior (Table 10). At 450M, the weak caution Packet remains at 0.500 with initialization 42 and order 1042, while initialization 1042 and order 42 reaches 1.000. At 27B, both crossed runs reach 1.000 on the photographic caution evaluation. These few interventions narrow the failure cases. They do not justify a general decomposition of initialization, optimization, and model-size effects. The complete per-seed rendered results, including the shorter mixed and doubled photo-only recipes, appear in Appendix E.

9 Limitations and implications

The packet can carry an operator mode at 27B, and a small detector can usefully decide when to request a new action in the tested stream. Those conclusions survive the additional controls. Their scope is a finite action task with known labels, fixed prompts, a small number of adapter seeds, and reused image sets.

Several experiments are needed before making broader claims. Unrestricted generation and instruction following need tasks whose outputs are more than three action labels. Paraphrase robustness needs additional phrases crossed with the same images, rather than two phrases assigned to alternating subsets. New image collections and new human identities would test domain generalization. A live stream with task-relevant changes, measured complete-system latency and energy, and an action-level failure criterion would be required for deployment claims. These are extensions of the present claims, not measurements that this paper already supplies.

The comparisons also leave mechanisms unresolved. The main Text arm uses a constant adapter; auxiliary losses and training random draws differ between adapter types. The multi-scale comparison changes model family as well as size. The box projection and blank-image studies cover only the two LFM hosts. Mixed-domain follow-ups were chosen after observing failures and do not include every seed and recipe combination. The bootstrap intervals cannot remove those design limitations.

For this architecture, the practical lessons are to report raw action shares alongside baseline differences, verify detector errors on the images where they occur, distinguish agreement from correctness, and retain unresolved response delays. Conditioning strength alone is not a safety mechanism.

10 Reproducibility and evidence

The public reproduction package contains 232 original JSON records, the study-specific experiment harness, input and checkpoint identities, CPU analysis, and this manuscript's renderer:

`https://github.com/kortexa-ai/legolm.paper/tree/main/reproductions/lookout-captain`

The analysis verifies evidence-file hashes, recomputes 22490 reported scalar quantities from 143 individual-image result records, replays their parsers, and evaluates 36 host/mode/gate combinations. The latter contain repeated measurements of the same episodes, not 36 independent datasets. Each table and figure is generated from the checked analysis. Original readings remain available alongside the corrected presentation.

The release distinguishes exact reproduction of saved-data analysis from repetition of training and inference. All 72 referenced adapter, projection, classifier, and detector files were found and matched their recorded SHA-256 identities in the publication audit. The 27B host revision was recorded in the experiment. The LFM records name models without immutable revisions; the release pins the cached snapshots recovered from the study machine and labels their provenance accordingly. The original records also lack a complete environment lock and hashes of any uncommitted source at execution time. A pinned public harness and recovered checkpoints make the study inspectable, but do not retroactively supply missing historical provenance or promise bitwise retraining.

The original temporary rendered PNG files were not recovered. The evaluation records retain the scene draws, image IDs, and both sets of pixel hashes. Reproduction commands regenerate the scenes and check those identities; a different rendering device can produce a different pixel version. Saved-response analysis requires neither these PNGs nor a GPU.

The HSRD-100 asset is attributed to Digital Reality Lab and used under CC BY 4.0. The package identifies the exact archive and its hash. External image, model, detector, and mesh licenses remain with their respective sources.

11 Conclusion

Mode packets change action selection in frozen vision-language models from 450M to 27B, with effects that depend on mode, adapter seed, and instruction phrase. Detector-controlled selection and event-triggered invocation can preserve much of that behavior in a restricted replay while avoiding most model calls. The same experiments reveal false actions, phrase-specific failures, unstable baselines, and long or unresolved invocation delays. Those observations define both the demonstrated result and the work needed for a broader multimodal or robotic system.

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A Conditioning controls and seed variation

Table 6: Wrong-mode control, absent-image target-action share, and per-seed Packet–Constant contrasts. Seed order is 42, 1042, 2042. For explore, the absent target action is proceed and a share of one is correct. For social and caution, a nonzero absent share is an incorrect greet or stop.

Host	Mode	Packet–Swapped	Packet on absent	Per-seed Packet–Constant
450M	Social	0.633 [0.500, 0.767]	0.011 [0.000, 0.033]	0.133/0.667/0.200
450M	Caution	0.678 [0.556, 0.800]	0.033 [0.000, 0.078]	0.967/0.400/0.967
450M	Explore	0.522 [0.456, 0.589]	1.000 [1.000, 1.000]	-0.033/0.333/-0.167
3B	Social	0.489 [0.322, 0.667]	0.044 [0.011, 0.089]	0.133/-0.233/0.167
3B	Caution	0.756 [0.711, 0.811]	0.022 [0.000, 0.056]	0.133/0.900/0.967
3B	Explore	0.800 [0.722, 0.867]	1.000 [1.000, 1.000]	0.500/0.833/0.333
27B	Social	0.978 [0.944, 1.000]	0.000 [0.000, 0.000]	0.433/0.533/0.967
27B	Caution	1.000 [1.000, 1.000]	0.044 [0.011, 0.089]	0.767/0.833/1.000
27B	Explore	0.989 [0.967, 1.000]	1.000 [1.000, 1.000]	0.800/0.600/0.000

B Contrary text and box controls

Table 7: Contrary-text comparisons on the initial 30 present and 30 absent photographs, averaged over three seeds. P: Packet; T^-: contrary text with the constant adapter. The paired difference is measured on present images. Absent P is a false action for social/caution and correct proceed for explore.

Host	Mode	Dose	P	T^-	$P + T^-$	$(P + T^-) - T^-$	Absent P
450M	Social	0.15	0.633	0.289	0.556	0.267 [0.067, 0.467]	0.011
450M	Caution	0.15	1.000	0.178	0.989	0.811 [0.722, 0.889]	0.033
450M	Explore	0.15	0.522	0.022	0.333	0.311 [0.222, 0.400]	1.000
450M	Social	0.225	0.500	0.111	0.489	0.378 [0.167, 0.589]	0.033
450M	Caution	0.225	1.000	0.267	1.000	0.733 [0.689, 0.789]	0.078
450M	Explore	0.225	0.467	0.467	0.489	0.022 [-0.122, 0.167]	1.000
3B	Social	0.15	0.489	0.033	0.278	0.244 [0.133, 0.367]	0.044
3B	Caution	0.15	0.756	0.000	0.411	0.411 [0.322, 0.500]	0.022
3B	Explore	0.15	1.000	0.056	0.922	0.867 [0.778, 0.944]	1.000
3B	Social	0.225	0.478	0.644	0.500	-0.144 [-0.344, 0.056]	0.133
3B	Caution	0.225	0.889	0.011	0.744	0.733 [0.678, 0.789]	0.178
3B	Explore	0.225	1.000	0.011	1.000	0.989 [0.967, 1.000]	1.000
27B	Social	0.15	0.978	0.222	0.767	0.544 [0.456, 0.644]	0.000
27B	Caution	0.15	1.000	0.111	0.889	0.778 [0.711, 0.844]	0.044
27B	Explore	0.15	1.000	0.289	0.978	0.689 [0.600, 0.778]	1.000
27B	Social	0.225	0.989	0.333	0.889	0.556 [0.467, 0.644]	0.000
27B	Caution	0.225	1.000	0.022	0.911	0.889 [0.833, 0.944]	0.033
27B	Explore	0.225	1.000	0.622	1.000	0.378 [0.300, 0.456]	1.000

C Invocation controls and censoring

The historical lag records are preserved. The corrected analysis reconstructs cut-to-next-invocation events directly from the step logs. If several cuts precede a single invocation, each cut receives its own delay. A cut on the last step without an invocation is censored at duration zero. Such a record does not mean the model responded immediately. A descriptive survival curve is included in the machine-readable analysis; cuts within an episode are dependent, so it is not used as an independent population-level survival test.

Table 8: Projection experiments on the 100 present photographs. Means use three seeds. Blank replaces the host image at projection training and evaluation; the detector still sees the photograph. Lying supplies zeros on present photographs. The full records also contain absent-image false-box responses.

Host image	Host	Mode	Switch	Boxes	Lying	Boxes–Switch
Visible	450M	Social	0.590	0.533	0.537	-0.057 [-0.100, -0.010]
Visible	450M	Caution	0.930	0.923	0.967	-0.007 [-0.050, 0.040]
Visible	3B	Social	0.443	0.453	0.457	0.010 [-0.010, 0.033]
Visible	3B	Caution	0.703	0.927	0.760	0.223 [0.187, 0.260]
Blank	450M	Social	0.000	0.087	0.000	0.087 [0.060, 0.117]
Blank	450M	Caution	0.000	0.160	0.000	0.160 [0.127, 0.193]
Blank	3B	Social	0.000	0.000	0.000	0.000 [0.000, 0.000]
Blank	3B	Caution	0.317	0.333	0.333	0.017 [0.003, 0.033]

Table 9: 27B gate controls. Agreement is averaged across three seeds. Wakes/cuts counts only the 167 post-start cuts with an observed next invocation; mean delay excludes censored cases and must be read alongside their count. Gate schedules are identical across hosts and seeds.

Gate	Calls avoided	Caution agree	Social agree	Resolved cuts	Mean delay	Censored
Every frame	0.000	1.000	1.000	167/167	0.00	0
Episode start	0.938	0.677	0.679	0/167	–	167
Verdict	0.835	0.990	0.981	130/167	1.44	37
Boxes	0.723	1.000	0.997	148/167	0.42	19
Pixels	0.053	1.000	0.999	167/167	0.00	0
Scene cuts	0.764	0.989	0.990	167/167	0.00	0

D Training-order diagnostics

Table 10: Caution on the 30 present photographic images under mixed training at 384 steps. Initialization and training-order seeds are crossed only in these selected runs. Both mode and constant adapters are retrained in each row.

Host	Initialization	Order	Constant	Packet
450M	42	42	0.567	0.500
450M	1042	1042	0.167	0.967
450M	42	1042	0.200	0.500
450M	1042	42	0.767	1.000
27B	42	42	0.467	0.500
27B	1042	1042	0.533	1.000
27B	42	1042	0.233	1.000
27B	1042	42	0.433	1.000

E Complete rendered-scene seed results

Each row below uses the same 100 present rendered scene draws. The recipe identifiers match the artifact filenames after `vision-mode-r3b-`. A bare host uses the original photographic adapters. `x2door` doubles photographic training to 384 steps; `mxdoor` adds rendered training at 192 steps; `mx2door` uses mixed training at 384 steps; `mx2ndoor` retains that mode adapter but uses the original constant baseline. `mx2o1042door` and `mx2o42door` change the training-order seed. Slashes separate results in the listed seed order.

Table 11: Per-seed rendered Packet shares and baseline contrasts.

Recipe–host	Mode	Seeds	Packet	Packet–Constant
27b	Social	42,1042,2042	0.930/0.940/0.000	0.840/0.940/0.000
27b	Caution	42,1042,2042	0.550/0.820/0.500	0.520/0.810/0.500
27b	Explore	42,1042,2042	1.000/1.000/1.000	0.120/0.010/0.000
3b	Social	42,1042,2042	0.020/0.080/0.080	0.020/0.080/0.080
3b	Caution	42,1042,2042	0.000/0.230/1.000	0.000/0.230/1.000
3b	Explore	42,1042,2042	1.000/1.000/1.000	0.000/0.000/0.000
450m	Social	42,1042,2042	0.500/0.410/0.490	0.500/0.410/0.490
450m	Caution	42,1042,2042	1.000/0.070/1.000	1.000/0.070/1.000
450m	Explore	42,1042,2042	1.000/1.000/0.500	0.000/0.000/-0.500
mx2door-27b	Social	42,1042,2042	1.000/1.000/0.940	0.790/0.150/0.930
mx2door-27b	Caution	42,1042,2042	0.500/1.000/1.000	0.430/0.970/0.060
mx2door-27b	Explore	42,1042,2042	1.000/1.000/1.000	0.280/0.880/0.950
mx2door-3b	Social	42,1042	0.500/0.500	0.500/-0.500
mx2door-3b	Caution	42,1042	1.000/1.000	0.860/1.000
mx2door-3b	Explore	42,1042	1.000/1.000	0.140/1.000
mx2door-450m	Social	42,1042,2042	0.500/0.500/0.500	0.500/-0.290/0.500
mx2door-450m	Caution	42,1042,2042	0.500/1.000/1.000	0.360/0.940/0.440
mx2door-450m	Explore	42,1042,2042	0.000/0.000/0.500	-0.860/-0.150/0.060
mx2ndoor-27b	Social	42,1042,2042	1.000/1.000/0.940	0.910/1.000/0.940
mx2ndoor-27b	Caution	42,1042,2042	0.500/1.000/1.000	0.470/0.990/1.000
mx2ndoor-27b	Explore	42,1042,2042	1.000/1.000/1.000	0.120/0.010/0.000
mx2ndoor-450m	Social	42,1042,2042	0.500/0.500/0.500	0.500/0.500/0.500
mx2ndoor-450m	Caution	42,1042,2042	0.500/1.000/1.000	0.500/1.000/1.000
mx2ndoor-450m	Explore	42,1042,2042	0.000/0.000/0.500	-1.000/-1.000/-0.500
mx2o1042door-27b	Social	42	1.000	0.230
mx2o1042door-27b	Caution	42	1.000	0.990
mx2o1042door-27b	Explore	42	1.000	0.780
mx2o1042door-450m	Social	42	0.500	-0.350
mx2o1042door-450m	Caution	42	0.500	0.410
mx2o1042door-450m	Explore	42	0.500	0.440
mx2o42door-27b	Social	1042	1.000	1.000
mx2o42door-27b	Caution	1042	1.000	0.980
mx2o42door-27b	Explore	1042	1.000	0.020
mx2o42door-450m	Social	1042	0.500	0.500
mx2o42door-450m	Caution	1042	1.000	0.010
mx2o42door-450m	Explore	1042	0.490	0.480
mxdoor-3b	Social	42,1042	0.500/0.430	0.480/-0.430
mxdoor-3b	Caution	42,1042	1.000/1.000	0.950/0.860
mxdoor-3b	Explore	42,1042	1.000/1.000	0.070/1.000
mxdoor-450m	Social	42,1042,2042	0.500/0.500/0.500	0.140/0.290/0.500
mxdoor-450m	Caution	42,1042,2042	0.500/1.000/1.000	0.160/0.210/0.360
mxdoor-450m	Explore	42,1042,2042	0.000/0.060/0.500	-0.300/0.060/0.140
x2door-3b	Social	42,1042	0.030/0.020	0.030/0.020
x2door-3b	Caution	42,1042	0.310/0.180	0.310/0.180
x2door-3b	Explore	42,1042	1.000/1.000	0.000/0.000